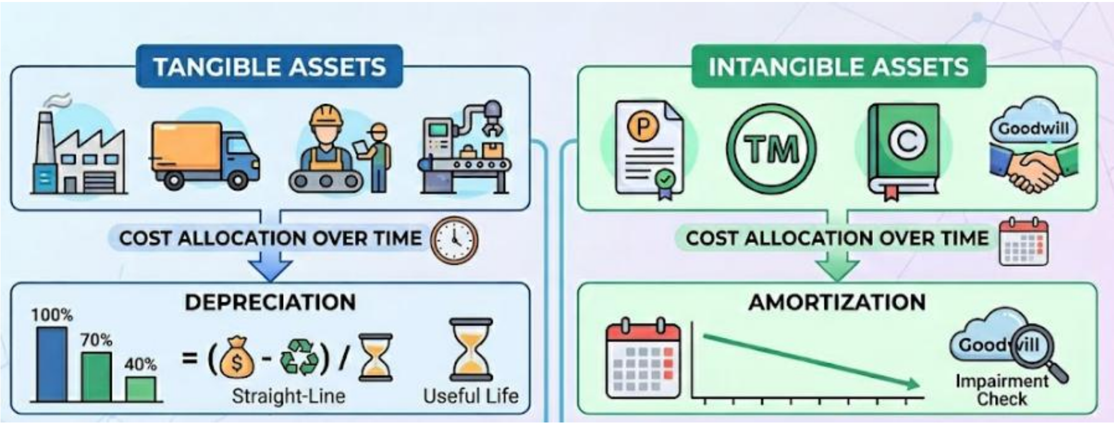
13. Long-Term Operating Assets (Ch 8)



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Part 1: Property,Plant, and Equipment (PPE)

13.0 Introduction

The 3 Key PPE Accounting Questions:

Because PPE is usually the largest and longest-lived asset category, accountants must answer:

1. **Capitalize vs. Expense** — Which costs go on the balance sheet as assets, and which hit the income statement immediately?
2. **Cost Allocation (Depreciation)** — How do we spread the capitalized cost across the periods that benefit?
3. **Disposals & Impairments** — How do we report asset sales or declines in fair value?

13.1 Tangible vs. Intangible Long-term Operating Assets

Long-term operating assets are used in operations for more than one period (not held for resale).

- **Tangible assets** — have physical substance (PPE: land, buildings, machinery, equipment).
- **Intangible assets** — lack physical substance but grant rights/privileges (patents, trademarks, copyrights, franchise rights, goodwill).

Two common traits: (1) used to generate revenue, not resold; (2) benefit multiple accounting periods.

EXHIBIT 8.1 Procter & Gamble Balance Sheet (assets only)

	June 30	
($ millions)	**2024**	**2023**
Assets		
Current assets		
Cash and cash equivalents	$ 9,482	$ 8,246
Accounts receivable	6,118	5,471
Inventories		
Materials and supplies	1,617	1,863
Work in process	929	956
Finished goods	4,470	4,254
Total inventories	7,016	7,073
Prepaid expenses and other current assets	2,095	1,858
Total current assets	24,709	22,648
Property, plant, and equipment, net	22,152	21,909
Goodwill	40,303	40,659
Trademarks and other intangible assets, net	22,047	23,783
Other noncurrent assets	13,158	11,830
Total assets	$122,370	$120,829

Concept Quiz 1 – Asset Classification

Fill in the blank with the correct balance sheet classification (**Inventory**, **PP&E**, or **Non-current Asset Held for Sale**).

1. A truck sitting on the lot of a car dealership, waiting to be sold to a customer, should be reported as ________________.

1) Inventory 2) PP&E 3) Non-current asset

2. A truck used by a delivery company to transport packages to customers should be reported as ________________.

1) Inventory 2) PP&E 3) Non-current asset

3. A truck that has been retired from delivery operations and is now parked, awaiting resale, should be reported as ________________.

1) Inventory 2) PP&E 3) Non-current asset

13.2 Determining Costs to Capitalize

- Capital expenditure vs. Revenue expenditure
  - **Capitalize** = record the cost as an asset on the balance sheet (to be depreciated/amortized later).
  - **Expense** = record the cost immediately on the income statement (reduces current-year net income).

- **Two Requirements to Capitalize**
  1. The asset is **owned or controlled** by the company.
  2. The asset is **expected to provide future benefits**.

- **What Costs Get Capitalized?**
  - **All "normal" costs necessary to acquire the asset AND prepare it for its intended use:**
    - Purchase price
    - Installation, taxes, shipping, legal fees, setup/calibration
    - **End-of-life legal obligations** (e.g., environmental cleanup, asset removal) — capitalized into the asset cost AND recorded as a liability at acquisition

- **Two Judgment Guardrails**
  1. **Direct linkage rule** — Only costs *directly linked* to future benefits can be capitalized. Incidental costs (those incurred whether or not the asset is purchased) → expense.
  2. **Future benefit ceiling** — Capitalized cost cannot exceed expected future cash inflows. (A $200 asset on the books should generate at least $200 in future cash flows.)

- **Self-Constructed Assets**
  - When a company builds an asset for its own use, capitalize:
    - All construction costs: **materials + labor + reasonable overhead**
    - **Capitalized interest** — interest expense incurred *during the construction period*

Concept Quiz 2 – What to capitalize?

1. ABC Corp. purchases a new manufacturing machine for $100,000, and incurs the additional costs listed below. Determine the total amount that should be capitalized as the cost of the machine?

Cost Item	Amount	Treatment	Why?
Purchase price	$100,000		
Shipping (freight-in)	$3,000		
Installation & calibration	$5,000		
Sales tax	$7,000		
CEO's salary for the month	$20,000		
Annual office rent	$12,000		
Employee training to use the machine	$4,000		

2. XYZ Bakery purchases a specialty oven for $50,000. However, given the oven's limited capacity and current bread prices, management realistically estimates that the oven will generate only $35,000 in expected future cash inflows over its useful life. **Record this transaction.** That is, determine (a) the amount to be capitalized as the cost of the oven, and (b) the impact on the income statement.

13.3 Depreciation – Concept and Methods

- **Definition**: Systematic allocation of a long-term asset's cost to each accounting period that benefits from its use.

- **Why needed**: Long-term operating assets benefit multiple periods, so their cost cannot be matched *directly* to revenues of a single period — accounting principles require an equitable allocation over the asset's useful economic life.

- **Effect on financial statements**:
  - **Income Statement**: Recorded as **Depreciation Expense**.
  - **Balance Sheet**: Offsetting credit to **Accumulated Depreciation** (a contra-asset account, denoted "XA").

- **Book value (carrying value)** = Asset cost − Accumulated depreciation.

- **Important caveat**: Book value ≠ Fair value.
  - Depreciation is a *cost allocation*, **not** a measure of decline in market/fair value.
  - Fair value may decline more, less, or even *increase* in a given period.

Comparison of Depreciation Methods

Example (Dehning Co.): Buys delivery truck for $120,000; Useful life 5 years. Residual value is $20,000.

Method 1: Straight-Line (SL)

Same expense each year. Rate = 1/5 = 20%.

Yr	Rate	Dep. Expense	Book Value
0	—	—	**$120,000**
1	20%	$20,000	$100,000
2	20%	$20,000	$80,000
3	20%	$20,000	$60,000
4	20%	$20,000	$40,000
5	20%	$20,000	$20,000
Total	**100%**	**$100,000**	**N/A**

Balance Sheet — End of Year 2

Truck (at cost)	$120,000
Less: Accumulated Depreciation	(40,000)
Book Value	**$80,000**

Method 2: Sum-of-Years'-Digits (SYD)

Front-loaded. Denominator = 1+2+3+4+5 = 15.

Yr	Rate	Dep. Expense	Book Value
0	—	—	**$120,000**
1	33.3% (5/15)	$33,333	$86,667
2	26.7% (4/15)	$26,667	$60,000
3	20.0% (3/15)	$20,000	$40,000
4	13.3% (2/15)	$13,333	$26,667
5	6.7% (1/15)	$6,667	$20,000
Total	**100%**	**$100,000**	**N/A**

Balance Sheet — End of Year 2

Truck (at cost)	$120,000
Less: Accumulated Depreciation	(60,000)
Book Value	**$60,000**

- Key Takeaways
  - Both methods produce the SAME total depreciation ($100,000) and the SAME final book value.
  - They differ only in the TIMING of expense recognition.

- Depreciation for Tax Purposes (US):
  - Most US companies use straight-line for financial reporting, but an accelerated method (MACRS) for tax returns.
  - Why? Higher early-year depreciation → lower taxable income → lower current tax → cash savings now (a form of tax deferral).
  - This difference between book and tax depreciation creates deferred tax liabilities (FYI only)

Concept Quiz 3

1. On January 2, Lev Company purchases equipment for $95,000 with an estimated useful life of 5 years and estimated salvage of $10,000. Using the straight-line method, what is the annual depreciation expense?

   a) $17,000 b) $19,000 c) $21,000 d) $9,500

2. The purpose of recording periodic depreciation of long-term PPE assets is to:

   a) report declining asset values on the balance sheet

   b) allocate asset costs over the periods benefited by use of the assets

   c) account for costs to reflect the change in general price levels

   d) set aside funds to replace assets when economic usefulness expires

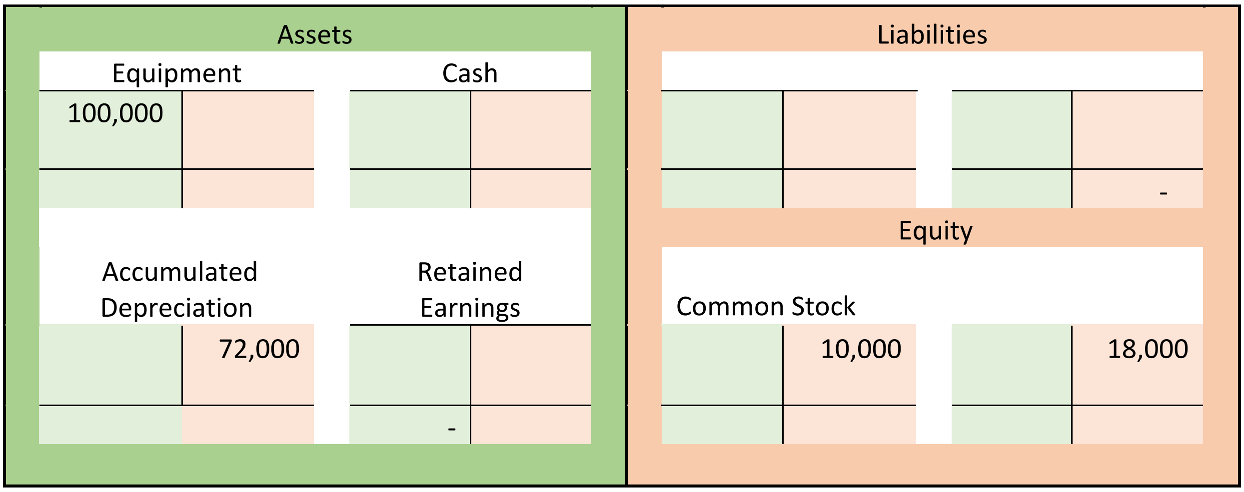
13.4 Assets Sales and Impairments

Gains and Losses on Asset Sales

- When a long-term asset is sold, both its original cost and its accumulated depreciation are removed from the balance sheet.
    - **Gain or Loss on Sale = Proceeds from Sale – Book Value of Asset Sold**
- Gain > 0 (Proceeds > BV) → recognized as income on the income statement
- Loss < 0 (Proceeds < BV) → recognized as a loss

Illustration – Sale of Dehning Delivery Truck

After 4 years of straight-line depreciation, Dehning decides to sell the truck for $30,000 cash. Book value at end of Year 4 = $100,000 – $72,000 = $28,000. What is the impact of this sale on the balance sheet and income statement? (i.e., journal entry)



- Before the sale: Assets (28,000) = Liabilities (0) + Stockholders' Equity (28,000)
- After the sale : Assets (30,000) = Liabilities (0) + Stockholders' Equity (30,000)

Journal Entry

Dr. Cash	$30,000	
Dr. Accumulated Depreciation	$72,000	
Cr. Truck (Asset – Original Cost)		$100,000
Cr. Gain on Sale of Truck (Revenue)		$2,000

Asset Impairments

- **General rule**: PPE is reported at **net book value** (cost − accumulated depreciation), even if fair value rises (unrecognized gains stay hidden).

- **When to recognize impairment**: Only if fair value has **permanently declined**.

- **Two-step impairment test (assets held for use) – FYI only**:
  1. **Recoverability test**: Compare *sum of expected (undiscounted) future cash flows* to **net book value**.
     - Cash flows > NBV → **No impairment**.
     - Cash flows < NBV → **Impaired** → proceed to Step 2.
  2. **Measurement**: Write asset down to **fair value** (typically the discounted PV of expected cash flows).

- **Effect on financial statements**:
  - **B/S**: Asset reduced by the write-down amount.
  - **I/S**: Impairment loss recognized → reduces net income (often classified within **restructuring costs**).

- **Going forward**: Future depreciation is **lower** because the written-down portion of cost is permanently removed from the balance sheet.

- **Link to Restructuring Charges**:

  - Impairment losses are often **bundled within restructuring costs** on the income statement, alongside severance and workforce reduction expenses.
  - Restructuring announcements (plant closures, product line exits) frequently trigger impairment testing by reducing expected cash flows.
  - The bundling amplifies management's **discretion over timing and amount**, making restructuring periods especially prone to the abuses below.

- **Management judgment & risks** (GAAP permits neither, but estimation latitude exists):
  - **Insufficient write-down**: Overstates current income; understates future income.
  - **Aggressive write-down ("big bath")**: Overstates future income by purging costs in one bad year.
- **Key reminder**: Impairment is a **noncash charge** for accounting purposes, but the cash *was* spent at original acquisition.
- **Note** →

- Notes related to PPE from P&G's 2024 Annual Report

The note details P&G's depreciation method (straight-line) and the estimated useful lives of various classes of PPE assets. Later in the notes, the company reports "asset-related costs" of $101 million included in its restructuring charges for the year ended June 30, 2024. The company describes these costs as follows:

> **Asset-related costs**
> Asset-related costs consist of both asset write-downs and accelerated depreciation for manufacturing and facilities consolidations. Asset write-downs relate to the establishment of a new fair value basis for assets held-for-sale or for disposal. These assets are written down to the lower of their current carrying basis or amounts expected to be realized upon disposal, less minor disposal costs. Charges for accelerated depreciation relate to long-lived assets that will be taken out of service prior to the end of their normal service period.

Illustration — Measuring Impairment Loss (FYI Only)

- ABC Co. owns a machine with a **net book value of $300,000** at year-end. Expected cash flows from continued use over the next 5 years are:

Year	1	2	3	4	5
Cash Flow	$70,000	$60,000	$55,000	$45,000	$30,000

The appropriate discount rate is **10%**.

Required:

a. Perform the recoverability test using undiscounted cash flows.
b. If impaired, measure the impairment loss using discounted cash flows (fair value).
c. Record the journal entry.

Solution

a. Recoverability Test — Undiscounted Cash Flows

$$Undiscounted\ CF = CF_1 + CF_2 + CF_3 + CF_4 + CF_5 = 70{,}000 + 60{,}000 + 55{,}000 + 45{,}000 + 30{,}000 = \$260{,}000$$

$260,000 < $300,000 NBV → **Asset is impaired.**

b. Measure Fair Value — Discounted Cash Flows (r = 10%)

$$Fair\ Value = CF_1/(1.10)^1 + CF_2/(1.10)^2 + CF_3/(1.10)^3 + CF_4/(1.10)^4 + CF_5/(1.10)^5$$

Year	Cash Flow	÷ $(1.10)^n$	Present Value
1	$70,000	÷ 1.1000	$63,636
2	$60,000	÷ 1.2100	$49,587
3	$55,000	÷ 1.3310	$41,322
4	$45,000	÷ 1.4641	$30,735
5	$30,000	÷ 1.6105	$18,628
	$260,000	**Fair Value**	**$203,908**

Impairment Loss = NBV − Fair Value = $300,000 − $203,908 = **$96,092**

c. Journal Entry

Account	Debit	Credit
Loss on Impairment	$96,092	
Accumulated Depreciation (or Equipment)		$96,092

- **Key Point:** Undiscounted cash flows ($260,000) determine *whether* to impair; discounted cash flows ($203,908) determine *how much*.

Concept Quiz 4

1. If the sale of a depreciable asset results in a loss, the proceeds from the sale were:
    a) less than current fair value
    b) greater than cost
    c) greater than book value
    d) less than book value

2. When is a PPE asset considered impaired under US GAAP?
    a) When fair value drops below book value
    b) When the sum of undiscounted expected future cash flows is less than book value
    c) When the company stops using it
    d) When the market value of similar assets falls

13.5 Asset Revaluation: US GAAP vs. IFRS (FYI)

One Rule Difference and One Policy Choice

Section 13.4 gave the US GAAP answer: write-downs are required, write-ups are prohibited. Under IFRS the answer differs.

- **A rule difference.** US GAAP forbids revaluing nonfinancial long-term assets upward. No election is available. This is a genuine framework difference.
- **A policy choice.** IAS 16 offers every IFRS filer a choice between the cost model (i.e., the US model) and the revaluation model, applied by class of asset. A filer on the cost model cannot write up — not because IFRS forbids it, but because that filer chose the model that does not permit it.
- **A transition mechanic.** IFRS 1 lets a first-time adopter measure an item of PP&E at fair value once, at the transition date, and treat that amount as deemed cost going forward. This is not the revaluation model and does not commit the company to ongoing remeasurement.

Three-Way Comparison

	US GAAP	IFRS — revaluation model	K-IFRS — deemed cost at transition
Authority	ASC 360 / ASC 350	IAS 16	IFRS 1 (K-IFRS 제1101호)
Write-up above original cost?	Prohibited	Permitted, no ceiling	Permitted once, at the transition date only
Where the credit goes	n/a	OCI — revaluation surplus in equity	Directly to retained earnings (opening equity)
Ongoing obligation	None	Revalue the whole class with sufficient regularity	None — reverts to the cost model immediately after
Later decline is measured as	Impairment (undiscounted cash flow screen first)	Revaluation decrease to fair value	Impairment under IAS 36 (recoverable amount)
Later recovery	Never recognized	Revaluation increase, no ceiling	Impairment reversal, capped at deemed cost
Typical user	All US filers	A minority of IFRS filers, most often for land	Many Korean filers at 2011 adoption

Do Not Confuse Impairment Reversal with Revaluation

	Impairment reversal (IAS 36)	Revaluation model (IAS 16)
What it does	Undoes a prior write-down	Marks the asset to fair value, up or down
Ceiling	Capped at depreciated historical (or deemed) cost	None — may exceed original cost
Gain recorded in	Profit or loss	OCI, except to the extent it reverses a prior P&L loss
Elective?	No — required when indicators reverse	Yes — an accounting policy choice by class

Key Point: Only the revaluation model can carry an asset above original cost. An impairment reversal alone can never get you there — it is capped at what the carrying amount would have been had the impairment never been recognized.

Illustration — Three Years, Side by Side (FYI Only)

- A parcel of land was acquired years ago for **$1,000**. Land is used so that depreciation does not obscure the comparison. Year 1 is the IFRS transition date for the Korean filer.
- Fair value is **$1,400** at the end of Year 1, falls to **$900** at the end of Year 2, then recovers to **$1,200** at the end of Year 3.
- For the US filer, undiscounted future cash flows are $1,300 in every year — above carrying amount — so the recoverability test of Section 13.4 is passed and no impairment is recorded.
- Costs of disposal are ignored and value in use is not higher. Deferred taxes are ignored.

	US GAAP (cost)	IFRS revaluation model	K-IFRS deemed cost
Year 1 — fair value $1,400 (transition date)			
Ending carrying amount	**1,000**	**1,400**	**1,400**
Effect on profit or loss	—	—	—
Effect on OCI (revaluation surplus)	—	+400	—
Effect on retained earnings	—	—	+400 (IFRS 1 adj.)
Year 2 — fair value $900			
Ending carrying amount	**1,000**	**900**	**900**
Effect on profit or loss	— (screen passed)	(100)	(500) impairment
Effect on OCI (revaluation surplus)	—	(400) exhausted	—
Year 3 — fair value $1,200			
Ending carrying amount	**1,000**	**1,200**	**1,200**
Effect on profit or loss	—	+100 (reverses loss)	+300 reversal
Effect on OCI (revaluation surplus)	—	+200	—
Cumulative position after three years			
Carrying amount	**1,000**	**1,200**	**1,200**
Cumulative effect on profit or loss	0	0	(200)
Revaluation surplus balance	—	200	—
Total equity vs. the cost model	**0**	**+200**	**+200**

Journal Entries

Case A — US GAAP: no entries in any of the three years. The land remains at $1,000. A decline in fair value alone does not trigger impairment when the undiscounted cash flow test is passed.

Case B — IFRS revaluation model. Year 1 — increase of $400 to OCI:

Account	Debit	Credit
Land	400↑	
OCI — revaluation surplus		400↑

Year 2 — decrease of $500; the existing surplus absorbs the first $400:

Account	Debit	Credit
OCI — revaluation surplus	400↓	
Loss on revaluation (P&L)	100↓	
Land		500↓

Year 3 — increase of $300; the first $100 reverses the prior P&L loss:

Account	Debit	Credit
Land	300↑	
Gain on revaluation (P&L)		100↑
OCI — revaluation surplus		200↑

Case C — K-IFRS deemed cost at transition. Year 1 — one-time step-up to fair value as deemed cost, taken directly to opening retained earnings, not through income or OCI:

Account	Debit	Credit
Land	400↑	
Retained earnings (IFRS 1 transition adj.)		400↑

Year 2 — the company is now on the cost model, so the decline is tested as an impairment under IAS 36 against the $1,400 deemed cost:

Account	Debit	Credit
Impairment loss (P&L)	500↑	
Land		500↓

Year 3 — recovery is an impairment reversal, capped at the $1,400 deemed cost:

Account	Debit	Credit
Land	300↑	
Reversal of impairment loss (P&L)		300↓

Key Point: Cases B and C end at the same carrying amount ($1,200) and the same increase in total equity (+$200), but the income statements differ sharply: the revaluation model ran a net zero through profit or loss, while the deemed cost filer absorbed a net $200 charge. Same balance sheet, different income statement.

Why This Mattered in Korea

- K-IFRS adopts IAS 16 unchanged; what differed in 2011 was uptake, not the rule. With land carried at decades-stale K-GAAP cost, many firms chose the IFRS 1 deemed cost election over the revaluation model — the same equity and debt-to-equity improvement, without recurring appraisals.
- Practical implication: a missing revaluation surplus does not mean no step-up — deemed cost credits retained earnings instead, so two filers holding identical land may report $1,000 and $1,200 on the same date.

So Can a Korean Company Still Choose the Revaluation Model?

- **Yes.** The 2011 write-ups came bundled with IFRS adoption, so the option looks expired. Only one of the three mechanisms below actually is.
- **Deemed cost was a one-time road.** The revaluation model was never closed and is not closed now.
- **A 2011 deemed cost filer can still switch today** — it simply takes on the ongoing obligations below.
- **Not limited to IFRS filers.** Unlisted Korean companies under 일반기업회계기준 also choose between the cost and revaluation models (permitted since 2008).
- **Listed or unlisted, a Korean company may carry land and buildings above historical cost** — a structural difference from the US, where no framework for a commercial enterprise permits it.

Mechanism	What it does	Available to a Korean filer today?
Revaluation model (IAS 16 / K-IFRS 1016)	Carries the asset at fair value on an ongoing basis; increases go to OCI as revaluation surplus, with no ceiling	Yes — may be adopted at any time as a change in accounting policy
Deemed cost (IFRS 1 / K-IFRS 1101)	One-time measurement at fair value on the transition date, credited to retained earnings, then back to the cost model	No — spent at 2011 for existing filers; available only to a first-time adopter
Impairment reversal (IAS 36 / K-IFRS 1036)	Reverses a prior write-down only, capped at the carrying amount that would have existed without it	Yes — required whenever the indicators reverse, independent of the model chosen

The Election Is Not Free

- If write-ups improve the balance sheet and Korean firms may make them, why do most stay on the cost model? Five costs, which together outweigh the one-time cosmetic gain.

  - **Class-wide and recurring** — no cherry-picking; appraisals every year, not once.
  - **A one-way door** — IAS 8 makes returning to cost nearly impossible.
  - **Higher depreciation forever after** — which is why Korean revaluations are almost all land.
  - **Downside hits income once the surplus is exhausted** — the $100 loss in Case B, Year 2.
  - **No tax benefit, plus an immediate deferred tax liability**. Korean tax law doesn't recognize the write-up, so the tax base is unchanged and nothing extra is deductible. The gap creates a taxable temporary difference under IAS 12, charged to OCI — so the surplus in equity is net of tax, not gross.

Illustration: a $400 revaluation increase at a 25% tax rate produces a $300 revaluation surplus in equity and a $100 deferred tax liability. The company has increased reported equity by $300, not $400, and has recognized a liability it will never pay in cash unless the asset is sold. The mechanics are the temporary-difference logic of Chapter 10, running in the opposite direction from the deferred tax assets discussed there.

What the Election Does to the Ratios

Part 3 of this chapter analyzes long-term assets through PPE turnover. A revaluation moves several of those measures at once, in opposite directions, without any change in the underlying business.

Measure	Effect	Why
Debt-to-equity ratio	Improves	Equity rises by the after-tax surplus; debt is unchanged. This is the effect Korean firms were pursuing in 2011.
Total assets / book value	Rises	The written-up carrying amount replaces stale historical cost.
Return on assets (ROA)	Worsens	A larger denominator, and for depreciable assets a smaller numerator as depreciation rises.
PPE turnover	Worsens	Sales are unchanged while average net PP&E increases — the ratio falls though nothing about efficiency changed.
Operating income	Falls (if depreciable)	Higher depreciation on the revalued base. No effect for land.
Operating cash flow	No effect	Revaluation is entirely noncash; only the deferred tax note and the equity section change.

Key Point: A Korean company can write its land up; the standard permits it and always has. The reason most do not is that the election buys a stronger balance sheet today at the price of weaker reported earnings and weaker asset-efficiency ratios for the rest of the asset's life — and it cannot be undone. When comparing a Korean filer with a US competitor, the first question is not which company performed better, but which measurement basis each one is using.

Concept Quiz 5 — Asset Revaluation

1. An IFRS filer carries a building under the cost model. Fair value is now well above cost. Can it write the building up?
    - a) Yes — IFRS always permits upward revaluation
    - b) No — IFRS prohibits carrying any asset above original cost
    - c) Not while on the cost model; it must first adopt the revaluation model for the entire class
    - d) Only up to the amount of a previously recognized impairment

2. Under the revaluation model, an asset with an existing $150 revaluation surplus declines in fair value by $220. How is the decline recorded?
    - a) $220 to profit or loss
    - b) $150 to OCI, $70 to profit or loss
    - c) $220 to OCI
    - d) $70 to OCI, $150 to profit or loss

Part 2: Intangible Assets

13.6 Nature of Intangible Assets

- **Intangible assets** = assets that lack physical substance but provide future benefits through specific property rights or legal rights.
  - Examples: patents, trademarks, copyrights, franchise rights, license agreements, customer lists, goodwill.

- The issues involved in reporting intangible assets are conceptually similar to those of accounting for PP&E with their own unique characteristics.

- Accounting challenges for intangible assets:
  - Benefits provided by intangible assets are often uncertain and difficult to quantify.
  - Useful life of intangible assets is also difficult to estimate with confidence.

Capitalization Rule — Purchased vs. Internally Developed:

Source	Accounting Treatment
• Intangibles PURCHASED from another party	• CAPITALIZE the purchase price → asset on balance sheet
• Intangibles developed INTERNALLY (R&D costs, internal trademark development, etc.)	• EXPENSE as incurred → not on balance sheet (with limited exceptions like legal fees to register an internally developed patent)

- *This asymmetry is one of the most-criticized features of US GAAP.*
- *A company that built its brand internally (Nike's swoosh) reports a much smaller intangible balance than one that acquires the same brand by purchase.*

Research and Development (R&D) Costs

- **US GAAP rule**: ALL R&D costs are **expensed as incurred**.
- **Rationale**: Future benefits from R&D are too uncertain to capitalize as an asset.[1]
- **R&D plant assets**:
  - Used **solely** for R&D with no alternative future use → **expensed** immediately.
  - With **alternative future uses** (general lab equipment, multi-purpose buildings) → **capitalized and depreciated** normally.

IFRS Comparison:

- **Research costs** – expensed (same as US GAAP).
- **Development costs** – **capitalized** if specific criteria are met (technical feasibility, intent to complete, ability to use/sell, probable future benefit, reliable cost measurement).

[1] Per the second capitalization rule ("the costs capitalized cannot exceed expected future benefits")

R&D Intensity – Examples (R&D / Sales):

Company	2020	2019	2018
Cisco Systems, Inc.	12.87%	12.67%	12.84%
Juniper Networks, Inc.	21.56%	21.50%	21.59%
HP, Inc.	2.61%	2.55%	2.40%

- High-tech firms invest 10%+ of revenue in R&D; consumer hardware firms much less.
- Because US GAAP forces all of this to be expensed, R&D-heavy firms **have understated assets and equity**, which **inflates ROA (=Earnings/Assets) and ROE**.

RESEARCH INSIGHT

Research has provided evidence consistent with managers reducing R&D spending when trying to meet certain earnings targets or other earnings goals. Part of this is due to the accounting—research and development costs are expensed immediately, reducing reported earnings in the current period. Thus, although the research and development spending should provide better future performance, it harms current performance leading "myopic" managers to cut back.[2] Recent research also suggests that some firms do not disclose research spending even when it appears that they must have such costs. One theory is that these firms do not want to disclose their research and development spending in order to hide the extent of their costs from their competitors.[3]

Concept Quiz 6

1. Which of the following best describes the current method of accounting for research and development costs under US GAAP?
   a) Revenue recognition method b) Systematic and rational allocation
   c) Immediate recognition as an expense d) Income tax minimization

2. A company buys a **multi-purpose** lab building for $5,000,000 that will be used 60% for R&D and 40% for non-R&D activities. How should the cost be accounted for?
   a) Capitalize 100% and depreciate over useful life
   b) Expense 100% immediately
   c) Capitalize 40%, expense 60%
   d) Capitalize 60%, expense 40%

[2] See Bushee, Brian, "The Influence of Institutional Investors on Myopic R&D Investment Behavior." *Accounting Review*, 1998; and Sloan, Richard and P. Dechow. "Executive Incentives and the Horizon Problem: An Empirical Investigation," *Journal of Accounting and Economics*, 1991, for examples of research on this topic.

[3] See Koh, P.S., and D. M. Reeb. "Missing R&D," *Journal of Accounting and Economics*, 2015.

13.7 Patents, Copyrights, Trademarks, and Francise Rights

Intangible Asset	Description	Useful Life
Patent	Exclusive right to produce a product or use a technology, granted by government	Legal life = 20 years; useful life is often shorter
Copyright	Exclusive right of an author/composer/artist over creative works	Life of creator + 70 years (for individuals)
Trademark	Registered name, logo, slogan, or design associated with a product (Nike swoosh, Coca-Cola bottle shape)	Indefinite – renewable forever
Franchise Right	Contractual right to operate a business in a defined area, for a defined period	Definite contractual period

Capitalization Rules:

- **Purchased patent:** Purchase price capitalized.
- **Internally developed patent:** R&D costs expensed; ONLY legal costs and registration fees are capitalized.
- **Internally developed trademark:** Advertising and development costs are expensed; the trademark itself is NOT on the balance sheet.
- *This is why many of the world's most valuable trademarks (Coca-Cola, Apple, Nike) do NOT appear as significant assets on the company's books.*

13.8 Amortization and Impariment of Intanbilbes

- Amortization
  - Systematic allocation of an intangible asset's cost over its useful life
  - the intangible-asset equivalent of depreciation
  - Method: Straight-line method is used.
  - Applies only to intangibles with DEFINITE lives (patents, copyrights, franchise rights).
  - Intangibles with INDEFINITE lives (some trademarks, goodwill) are NOT amortized.

- Illustration – Landsman Patent

  Landsman Company spent $100,000 in early 2026 to purchase a patent. The patent had a remaining LEGAL life of 12 years, the USEFUL life is estimated at 5 years.

  (1) Capitalize the purchase (Year 2026):

Dr. Patent (asset)	$100,000	
Cr. Cash		$100,000

(2) Annual amortization at end of 2026:
Annual Amortization = $100,000 ÷ 5 years = $20,000

Dr. Amortization Expense	$20,000	
Cr. Patent		$20,000

- Impairment of Intangibles with Indefinite Lives
  - Tested annually
  - Similar to PP&E (i.e., tangible long-live assets)

Concept Quiz 7

1. Why are some intangible assets amortized while others are not? What is meant by an intangible asset with an "indefinite life"?

2. Bowen Company developed a new production process internally that significantly reduced manufacturing cost. R&D costs were $120,000. The process was patented on July 1, Year 8. Legal costs and fees to acquire the patent totaled $12,500. Bowen estimated the useful life at 10 years.

Required:

a) How should Bowen account for the $120,000 R&D costs?

b) How should Bowen account for the $12,500 legal costs?

c) What annual amortization expense should Bowen record in Year 8?

13.9 Goodwill

- **Goodwill** arises ONLY when one company acquires another, and the purchase price exceeds the fair value of the acquired company's identifiable net assets.
  - **Goodwill = Purchase Price – Fair Value of Identifiable Net Assets Acquired**
  - Internally generated goodwill is NEVER recognized on the balance sheet.

- **Illustration – P&G's $35.3 Billion Goodwill**
  - In 2006, P&G paid $53.4 billion to acquire The Gillette Company. P&G recognized: $35.3 billion as goodwill, $29.7 billion to other identifiable intangibles, with the remainder going to tangible assets and liabilities assumed.
  - *Most of that goodwill remains on P&G's balance sheet today (Goodwill = $39.9 billion in 2020).*

- Goodwill Impairment
  - Goodwill is NOT amortized (i.e., considered to have an indefinite-life ???).
  - Tested annually for impairment.
  - If the fair value of the acquired business (reporting unit) is less than its book value, goodwill is written down to an imputed value.
  - *Example: HP's $10 billion goodwill write-off in 2012 related to its acquisition of Autonomy was nearly the entire $11 billion purchase price.*
  - **Recoveries (write-ups) are NOT allowed for goodwill** — once impaired, always impaired.

Concept Quiz 8

1. Goodwill should be recorded in the balance sheet as an intangible asset only when:

   a) it is sold to another company

   b) it is acquired through the purchase of another business

   c) a company reports above-normal earnings for five or more consecutive years

   d) it can be established that a definite benefit has resulted from some item such as an excellent reputation for service

2. Why does Procter & Gamble report a goodwill balance of nearly $40 billion, while a similar consumer-products company that grew organically reports near-zero goodwill?

Part 3: Analyzing Long-Term Operating Assets

13.10 PPE Turnover and Percent Depreciated

Ratio 1: PPE Turnover (PPET)

- Gauges how effectively company uses its physical assets to generate sales.

PPE Turnover (PPET) = Sales Revenue ÷ Average PPE, net

- Interpretation: Sales generated per dollar of PPE. Higher is generally preferred — implies lower capital investment is needed to support a given level of sales.
- Varies dramatically by industry: capital-intensive businesses (airlines, telecom) have very LOW PPET; asset-light businesses (consulting, software) have very HIGH PPET.

Illustration – P&G PPE Turnover:

Year	Sales	Avg PPE Net	PPET
2019	$67,684	($20,600 + $21,271) / 2 = $20,936	3.23
2020	$70,950	($21,271 + $20,692) / 2 = $20,982	3.38

- *Comparison: Delta Air Lines (PPET ≈ 0.59) vs. Cisco Systems (PPET ≈ 18.61) — Cisco is far less capital-intensive than Delta.*

Ratio 2: Percent Depreciated

- *Gauges the age of a company's long-term tangible assets relative to their expected useful lives.*

Percent Depreciated = Accumulated Depreciation ÷ Cost of Depreciable Assets

- Interpretation: How "used up" the company's PPE is. A high ratio (close to 100%) means assets are near the end of their useful lives — major capex may be needed soon.
- A low ratio (close to 0%) suggests a young asset base.
- CAUTION: "Cost of depreciable assets" excludes land and construction in progress (neither is depreciated).

Illustration – P&G Percent Depreciated:

Year	Accumulated Depreciation	Cost of Depreciable Assets*	Percent Depreciated
2019	$22,122	$40,009	55.3%
2020	$23,079	$40,960	56.3%

- *Both P&G and Colgate-Palmolive run a steady-state 54%–59%, typical of mature companies that replace assets continuously.*

Other Cautions When Using These Ratios:

- Different depreciation methods (SL vs. accelerated) and different useful-life assumptions can make two otherwise similar companies look very different.
- Outsourcing reduces PPE in the numerator → higher PPET, but that's a strategic choice, not necessarily greater efficiency.

13.11 Discussion Questions

Q8-1. Maintenance vs. Improvement.

How should companies account for costs (maintenance or improvements) incurred after an asset is acquired?

Q8-2. Capitalized Interest.

What is the effect of capitalized interest on the income statement in the period an asset is constructed? What is the effect in future periods?

Q8-4. Accelerated Depreciation for Tax Purposes.

Why do companies use accelerated depreciation for income tax purposes, when the total depreciation taken over the asset's useful life is identical to straight-line depreciation?

Q8-8. R&D and GAAP.

What is the proper accounting treatment for research and development costs? Why are R&D costs NOT capitalized under GAAP?

Q8-9. Definite vs. Indefinite Life Intangibles.

Why are some intangible assets amortized while others are not? What is meant by an intangible asset with an "indefinite life"?

Q8-10. Goodwill Reporting and Impairment.

Under what circumstances should a company report goodwill on its balance sheet? What is the effect of goodwill on the income statement?

You Make the Call – Division Manager

You are the division manager of a main operating division. You are concerned that a declining PPE turnover (PPET) is adversely affecting your division's profitability. What specific actions can you take to increase PPET?

-- The End of Chapter 8 --